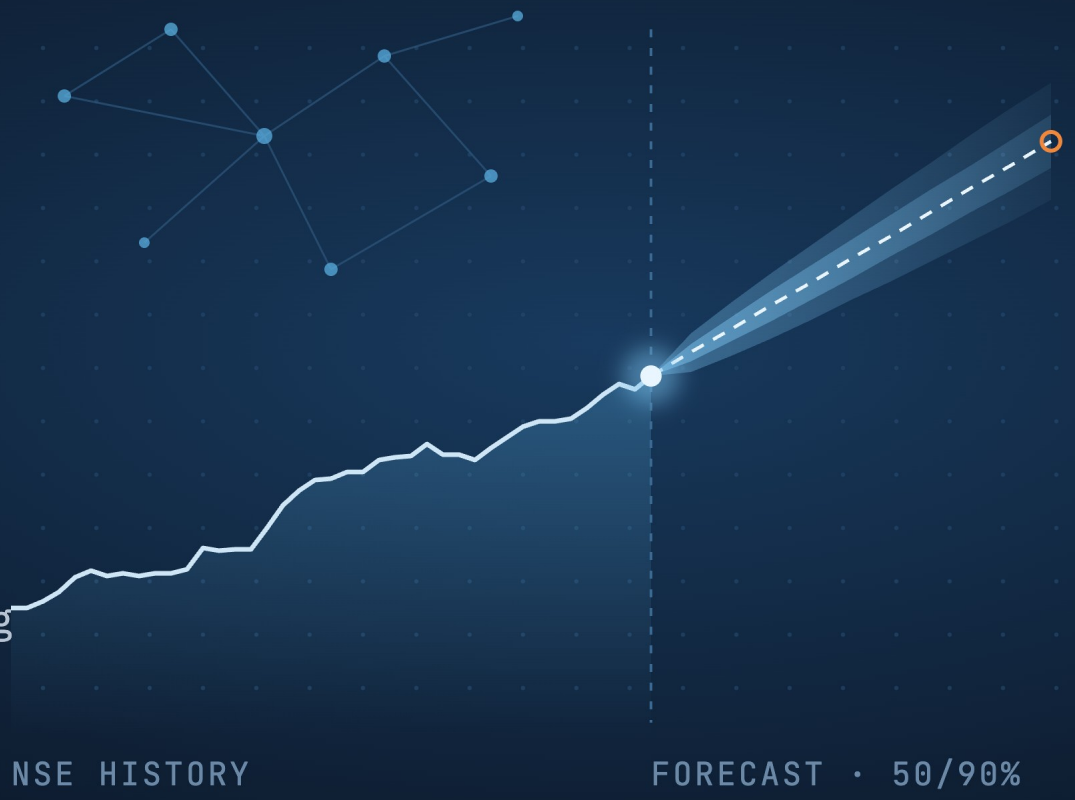
ALPHAEDGE
QUANT AND FINANCIAL ENGINEERING ACADEMY

CORPORATE PROGRAMME · 2026

ML · DEEP LEARNING · GENAI · AGENTIC AI

AI for the Trading Desk

From NSE market data to tested, explainable and governed trading algorithms — for teams who already know the market.



6
learning modules

6
NSE algorithms, built end to end

4
AI families: ML, DL, LLM, agents

SEBI
aware governance throughout

ABOUT ALPHAEDGE

Industry depth meets research rigour

ALPHAEDGE
QUANT AND FINANCIAL ENGINEERING ACADEMY

We train capital-market professionals to build, test and govern AI-driven trading systems.

- ✓ PhD and M.Tech-qualified faculty
- ✓ Practitioners in AI/ML, data and markets
- ✓ Built around NSE data and SEBI rules

AI, ML and deep learning

[Production models we have built, e.g., forecasting, NLP, risk scoring — with scale and outcome]

Data engineering at market scale

[Tick-data platforms, real-time pipelines, data volumes handled]

Indian capital markets

[Desks and products — equity, F&O, quant research — plus publications or patents]

WHY NOW

The Indian algo desk changed in 2026

1 Apr 2026

SEBI's framework for safer retail algo trading became fully applicable to all stock brokers.

Algo ID

Every algo order carries an exchange-issued tag. Above 10 orders per second per exchange, the strategy must be registered.

AI rules

SEBI has consulted on responsible AI/ML use and signalled human oversight and kill switches.

Desks that understand AI and its controls will build faster — and stay compliant.

Sources: SEBI circular, 30 Sep 2025 · NSE implementation standards, 5 May 2025 · SEBI AI/ML consultation, 20 Jun 2025

COURSE MAP

Six modules, one capstone

M1

Data foundation

NSE data, cleaning, features, backtest biases

M2

Machine learning

Classifiers, regimes, pairs, walk-forward tests

M3

Deep learning

LSTM, CNN and Transformers for time series

M4

GenAI and LLMs

Prompt engineering; filings and concalls at scale

M5

RAG, vector DB, MCP

AI connected to your documents and tools

M6

Agents and governance

AI teammates inside SEBI-aware guardrails

CAPSTONE Build, backtest and paper-trade an ML + LLM signal on real NSE data

THE MENTAL MODEL

AI vocabulary, translated into trading

AI term	What it means	Your trading equivalent
Model	A rulebook learned from data	Your strategy, with learned weights
Features	The inputs the model sees	Indicators: RSI, OI change, VWAP gap, PCR
Label	What we want to predict	Next-day direction, 5-day volatility
Training	Fitting the model to history	Building the strategy on past data
Validation	Testing on data it has never seen	Out-of-sample backtest, paper trading
Overfitting	Memorising noise, not signal	The curve-fitted backtest that fails live
Inference	Running the model on new data	Generating today’s live signal
Drift	Data no longer looks like history	Regime change, e.g., a volatility shock

THE MENTAL MODEL

Rule-based algo vs. learned algo

Rule-based: what you run today

```
IF RSI(14) < 30
AND close > SMA(200)
THEN BUY 1 lot
STOP 1.5 × ATR
```

Clear and simple. But 30, 200 and 1.5 are your guesses — and they stay fixed while the market changes.

Learned: what ML adds

```
X = [RSI, ATR, OI_chg, PCR, VIX,
FII_flow, … +40 more]
p_up = model.predict(X)
IF p_up > 0.60 → BUY, size ∝ edge
```

The model learns which signals matter, how much, and in what combination — from years of NSE history.

ML does not replace your judgment. It turns your judgment into a **testable, measurable** process.

THE MENTAL MODEL

The trading ML pipeline, end to end

RESEARCH · build the model

01 Collect
Bhavcopy, ticks, option chain, news

→

02 Clean
Splits, bonuses, holidays, gaps

→

03 Features
Returns, RSI, OI, VIX, flows

→

04 Label
What to predict, over what horizon

→

05 Train
Fit on the past only

PRODUCTION · earn the right to trade

06 Walk-forward
Retrain and test, window by window

→

07 Costs
STT, brokerage, GST, stamp, slippage

→

08 Paper trade
Live data, no money at risk

→

09 Go live
Small size, hard limits, kill switch

→

10 Monitor
Drift, decay, retrain triggers

MODULE 1 · DATA FOUNDATION

Six layers of Indian market data

End-of-day prices

NSE bhavcopy: OHLC, volume, delivery %. The base layer for daily models.

Intraday and tick

1-minute bars and tick-by-tick trades for intraday signals and execution.

Derivatives

Option chain, open interest, implied volatility, PCR, India VIX.

Results and events

Quarterly results, shareholding, corporate actions, NSE announcements.

Flows and macro

FII/DII activity, RBI policy, USD/INR, crude oil, GIFT Nifty.

Alternative text

News, concall transcripts, annual reports, SEBI circulars — fuel for LLMs.

Rule #1: every record is stamped with when it was actually available to you — not the date it describes.

Features: what the model actually sees

Feature	Simple formula	What it captures
1-day log return	ln(Close today ÷ Close yesterday)	Momentum, scale-free
20-day volatility	stdev(daily returns, 20) × √252	Risk regime
RSI(14)	100 − 100 ÷ (1 + avg gain ÷ avg loss)	Overbought / oversold
Gap %	(Open − prev close) ÷ prev close	Overnight news reaction
VWAP gap	(Close − VWAP) ÷ VWAP	Intraday buying pressure
OI change %	(OI today − OI yesterday) ÷ OI yesterday	New positions being built
Put-call ratio	Total put OI ÷ total call OI	Option-market positioning
IV percentile	Rank of today's IV within 1 year	Options cheap or rich?
FII flow, 5-day	Sum of FII (buy − sell), 5 days	Institutional direction

BEFORE ANY MODEL

Four ways a great backtest lies to you

Look-ahead bias

Using today's close to trade today's open, or results data before the announcement time.

Survivorship bias

Testing only on today's NIFTY 50 members ignores the stocks that were dropped along the way.

Data snooping

Try 500 parameter sets and keep the best one — some will win by luck alone.

Ignoring costs

STT, exchange charges, GST, stamp duty and slippage can erase a thin intraday edge.

Four families of ML, four trading jobs

Regression

Predicts a number.

ON THE DESK

Next-day return, 5-day volatility, fair spread

Classification

Predicts a category.

ON THE DESK

Up or down tomorrow; will this breakout hold?

Clustering

Finds groups, with no labels.

ON THE DESK

Market regimes; stocks that move together

Reinforcement

Learns by trial and reward.

ON THE DESK

Order execution, position sizing, hedging

Core models, in plain language

Model	How it thinks	NSE use case	Watch out for
Logistic regression	Weighted sum of signals	Baseline up/down classifier	Misses non-linear links
Random forest	Hundreds of trees vote	Will this breakout hold?	Can overfit noise
XGBoost / LightGBM	Trees that fix past errors	Next-day direction	Needs walk-forward tests
K-Means	Groups similar days	Calm, trend, stress regimes	You pick the group count
Hidden Markov model	Infers hidden market states	Strategy switching	Sensitive to set-up
PCA	Many inputs, few factors	Sector and risk factors	Factors are hard to name

Start simple. If logistic regression finds no edge, a deep network will usually find noise.

ALGORITHM 1 · CLASSIFICATION

NIFTY next-day direction classifier

01 **Data** — 10 years of NIFTY 50 daily bars, India VIX, FII/DII flows, GIFT Nifty gap.

02 **Label** — 1 if tomorrow's close beats today's by more than costs, else 0.

03 **Features** — 1/5/20-day returns, RSI, ATR, VIX change, PCR, 5-day FII flow.

04 **Train** — LightGBM, walk-forward: train on 3 years, test the next 3 months, roll.

05 **Signal** — long above 58% probability, short below 42%, otherwise flat.

06 **Risk** — size by volatility, daily loss limit, exit by the close.

```
for w in walk_forward(train="3y", test="3m"):
  m = LGBMClassifier(max_depth=4)
  m.fit(X[w.train], y[w.train])
  p = m.predict_proba(X[w.test])[:, 1]
  sig = where(p > 0.58, 1,
        where(p < 0.42, -1, 0))
  pnl += backtest(sig, costs=NSE_COSTS)
report(pnl) # Sharpe, max DD, hit rate
```

Illustrative parameters. Tune them inside walk-forward windows — never on the full history.

ALGORITHM 2 · UNSUPERVISED LEARNING

Market regime detection

INPUTS · DAILY

- 20-day return
- 20-day volatility
- India VIX level
- Breadth: % of stocks above 50-DMA

MODEL

Hidden Markov model or K-Means

3 states, retrained monthly

Calm trend

Momentum and breakouts, full size

Range-bound

Mean reversion, option selling

Stress

Cut size, hedge, or stay flat

Most strategies work in one regime and bleed in another. **The regime model decides which strategy may trade today.**

ALGORITHM 3 · STATISTICAL ARBITRAGE

Pairs trading with cointegration

01 **Pick** — two same-sector F&O stocks, e.g., two large private banks.

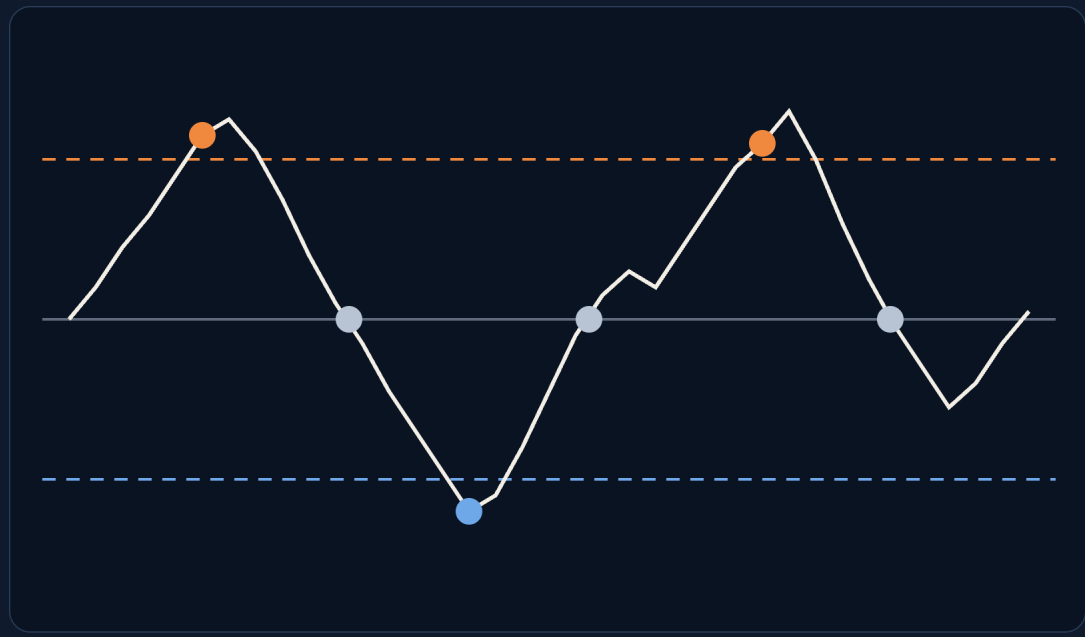
02 **Test** — cointegration on 1 year of data (Engle–Granger, p < 0.05).

03 **Hedge** — regress A on B to get β; spread = A − β × B.

04 **Score** — z = (spread − 60-day mean) ÷ 60-day std.

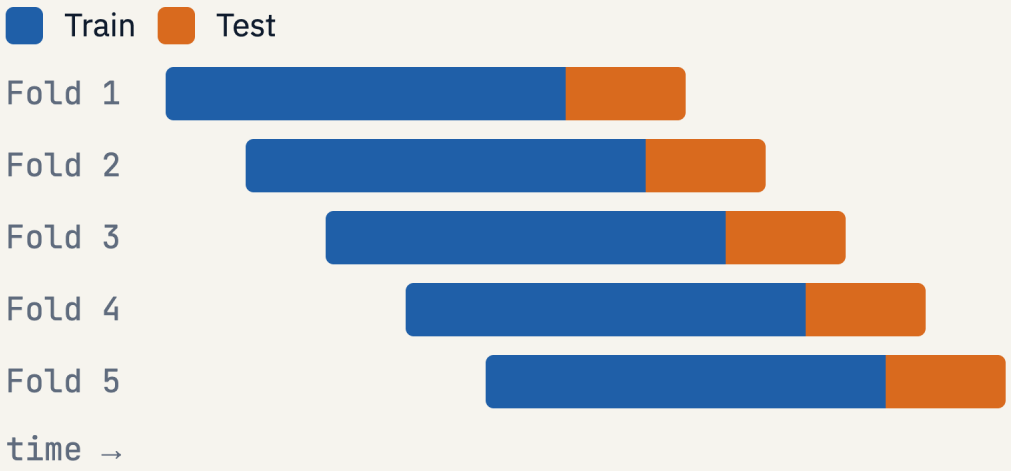
05 **Trade** — z above +2: short A, long B. Below −2: the reverse. Exit at 0.

06 **Protect** — stop beyond ±3.5 or after 20 days; re-test the pair monthly.



Orange dot: z crosses +2, short the spread · Blue dot: z crosses −2, long the spread · Grey dot: back to 0, exit

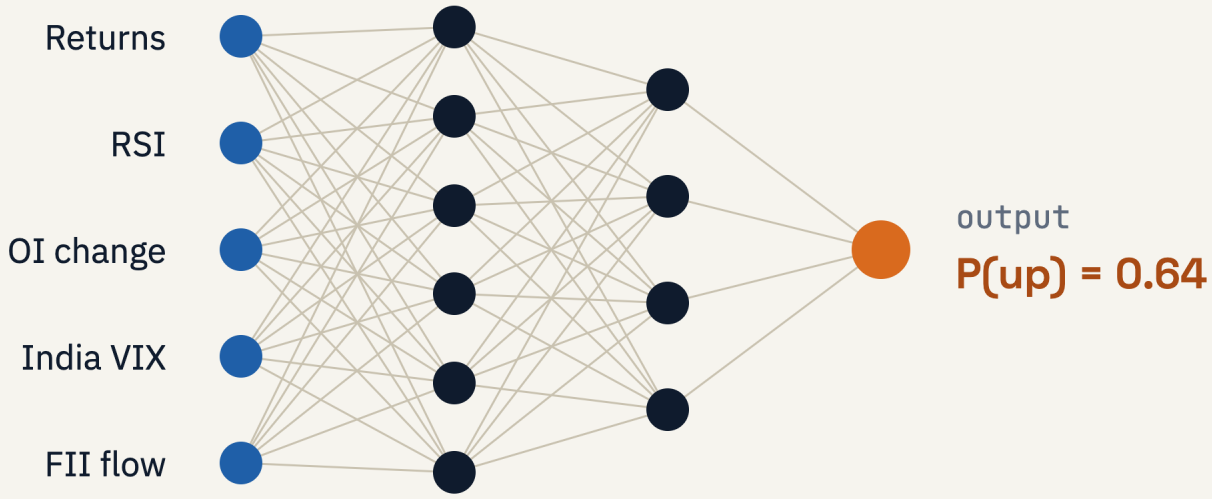
Proving the edge: walk-forward testing



The model never sees the test months while it learns. Then the window rolls forward — exactly as it would live.

Metric	Plain meaning	Rule of thumb
Sharpe ratio	Return per unit of risk	> 1, net
Sortino ratio	Return ÷ downside risk	> 1.5, net
Max drawdown	Worst fall from a peak	Survivable
Hit rate	Share of winning trades	Read with R:R
Profit factor	Profits ÷ losses	> 1.3
Calmar ratio	Yearly return ÷ max DD	> 1

Neural networks, in one picture



Each node is a tiny analyst. It weighs a few inputs and passes an opinion to the next layer.

Layers combine opinions into patterns that no single indicator shows.

Training nudges millions of weights until predictions on history are least wrong (backpropagation).

Use it where data is rich — ticks, order books, text. On a few thousand daily rows, simpler models often win.

Deep learning architectures for markets

LSTM / GRU

Reads prices as a sequence and remembers what mattered earlier.

Desk use: volatility and return forecasts from 60–120-day windows.

1D-CNN

Scans short windows for local shapes, like a chart reader.

Desk use: pattern detection on OHLC bars and order-book snapshots.

Transformer

“Attention” decides which past days, or which stocks, matter most today.

Desk use: multi-horizon forecasts (Temporal Fusion Transformer).

Autoencoder

Learns what “normal” looks like, then flags the unusual.

Desk use: abnormal OI build-up, volume spikes, risk alerts.

ALGORITHM 4 · DEEP LEARNING

LSTM volatility forecaster for options

INPUT · 60 DAYS

NIFTY returns, realised vol, India VIX, ATM IV, PCR

→

MODEL

LSTM, 2 layers × 64 units, dropout 0.2

→

OUTPUT

Forecast of next-5-day realised volatility

Decision rule

> **IV well above forecast** → options look rich → defined-risk selling, e.g., iron condor

> **IV well below forecast** → options look cheap → long straddle or strangle

Filters: skip RBI policy, Budget and results days; cap loss per trade.

```
model = Sequential([
  LSTM(64, return_sequences=True),
  Dropout(0.2),
  LSTM(64),
  Dense(1)  # 5-day realised vol
])
model.fit(X_60d, y_vol5d, epochs=50)
edge = atm_iv - model.predict(x_today)
```

What a large language model really is

An LLM predicts the next word, after reading a vast share of the world's text. That makes it a superb reader, writer and coder — not a price oracle.

Token
A word-piece. Cost and limits are counted in tokens.

Context
How much it can read at once — a full annual report, or more.

Prompt
Your instruction. Structure in, structure out.

Temperature
0 gives consistent answers; higher is more creative.

Hallucination
Confident but wrong. Fix it with sources and checks.

Where LLMs help the desk today

Read at scale

Summarise 200 concall transcripts overnight: guidance, capex, margins.

Event triage

Classify NSE announcements by type and materiality in seconds.

Sentiment features

Turn news and filings into numeric scores your ML models can use.

Code co-pilot

Write, explain and review Python for backtests and data pipelines.

Regulation helper

Explain SEBI and exchange circulars; draft checklists for sign-off.

Research drafts

Draft morning notes and post-trade reports from structured data.

Rule: LLMs read and write. Numbers, signals and orders come from tested code.

Same model, very different answers

Weak prompt

```
Summarise this concall.
```

- A different answer on every run
- Rounds or skips key numbers
- May invent guidance that was never given

Engineered prompt

```
ROLE: Banking analyst, Indian equities.
TASK: From the transcript, extract FY27
guidance: loan growth, NIM, credit cost.
RULES: Use only management's words.
If a figure is not stated, return null.
FORMAT: JSON {metric, value, quote}
TRANSCRIPT: """ … """
```

- Same structure on every run
- Every figure tied to a direct quote
- Says “null” instead of guessing

Anatomy of a production prompt

1 · Role

Who the model should act as.

`“Credit analyst covering PSU banks.”`

2 · Context

The data it must use — and only that.

`“Use only the Q2 FY27 transcript.”`

3 · Task

One clear, checkable job.

`“Extract NIM and slippage guidance.”`

4 · Constraints

What it must never do.

`“No forecasts. If missing, return null.”`

5 · Output format

A schema your code can parse.

`{"metric", "value", "quote"}`

6 · Examples

One or two solved cases (few-shot).

`Input: … → Output: {…}`

MODULE 4 · PROMPT ENGINEERING

Six techniques for financial work

Technique	What it does	Use it on the desk for
Zero-shot	Instruction only, no examples	Quick summaries, clear-cut classification
Few-shot	Adds 2–5 solved examples	Consistent labels, e.g., event type
Step-by-step reasoning	Reasons before it answers	Multi-step analysis, scenario logic
Structured output	Enforces a JSON schema	Anything that feeds a database or model
Prompt chaining	Splits a job into linked prompts	Extract → verify → summarise long reports
Self-check	A second prompt audits the first	Catching invented numbers early

PROMPTS ARE CODE

Version control
Every prompt lives in Git

Test sets
Known answers, re-run on every change

Temperature 0
Repeatable output for backtests

Re-test on upgrade
A new model version is a new strategy

ALGORITHM 5 · GENERATIVE AI

Announcement sentiment as a feature

1 COLLECT
NSE announcements and news, timestamped

→

2 SCORE
LLM returns JSON: type, sentiment, materiality

→

3 VALIDATE
Schema check; humans audit a sample

→

4 AGGREGATE
Per stock, per day, decay-weighted

→

5 FEED
New features into the Algorithm 1 model

```
SYSTEM: Indian equity analyst. JSON only.
USER: Classify this NSE announcement:
"<announcement text>"
→ {"type": "order_win",
   "sentiment": 0.7,
   "materiality": 4,
   "reason": "order ≈ 8% of revenue"}
```

Rules that keep it honest

- Use announcement time, not processing time
- Temperature 0; freeze the model version
- Keep it only if it adds out-of-sample value
- Log every prompt and output for audit

Embeddings: search by meaning

01 An embedding model turns any text into a vector — a list of hundreds or thousands of numbers.

02 Similar meaning lands close together, even with no shared words.

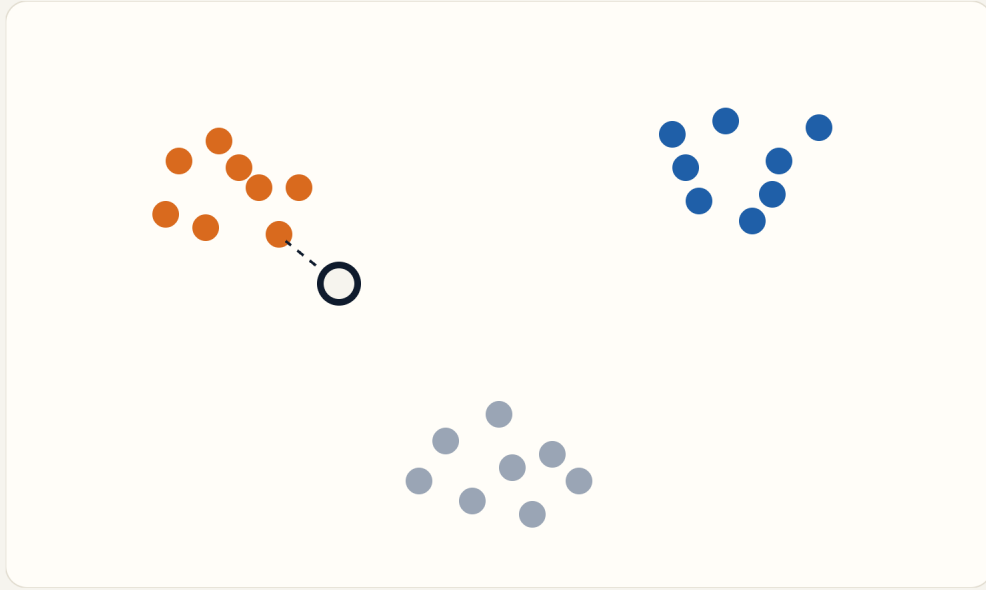
03 A vector database stores millions of them and finds the nearest in milliseconds.

EXAMPLE

Query: “management expects margin pressure”

Finds: “input-cost inflation will weigh on EBITDA for two quarters”

FAISS · pgvector · Qdrant · Milvus · Chroma



MODULE 5 · RETRIEVAL-AUGMENTED GENERATION

RAG: the LLM answers from your documents

INDEXING · once, then daily

Documents
SEBI circulars, reports, concalls

→

Chunk
Split into short passages

→

Embed
Each passage becomes a vector

→

Vector DB
Stored with source and date

ANSWERING · every question

Question
“Which banks guided lower slippages?”

→

Embed
Same model, same vector space

→

Retrieve
Top-5 nearest passages

→

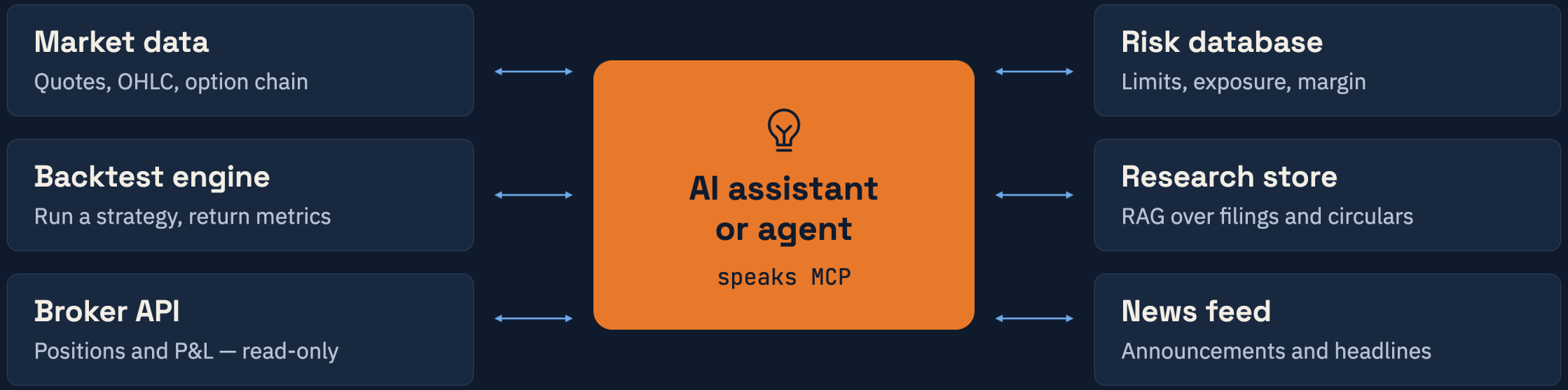
LLM answer
Written from passages, with citations

Why RAG: answers grounded in sources you trust, on private and current data — every claim traceable to a page.

MODULE 5 · MODEL CONTEXT PROTOCOL

MCP: one standard plug for every tool

An open standard that lets any AI app connect to your data and tools — think of it as USB-C for AI.



Build each tool once; any model or agent can use it. Orders stay behind human approval.

What makes AI “agentic”



Brain
An LLM that reasons and picks the next step

Tools
Reached through MCP: data, backtests, documents

Memory
Vector database plus a log of past steps

Guardrails
Limits, permissions and human approval

A multi-agent research desk

ORCHESTRATOR Splits the task, gathers results, writes the trade proposal

↓

Data
Pulls prices, OI and flows via MCP

Analyst
Reads filings and concalls via RAG

Quant
Runs backtests; reports Sharpe and drawdown

Risk
Checks exposure, margin and loss limits

Compliance
Checks the plan against SEBI and exchange rules

↓

HUMAN **Portfolio manager reviews, edits and approves — or rejects**

↓

EXECUTION Through the broker’s approved algo channel, tagged with its Algo ID

ALGORITHM 6 · AGENTIC AI

Pre-market brief agent, every trading day

07:30	**Collect**	GIFT Nifty, global indices, crude and USD/INR via MCP data servers
07:45	**Read**	Overnight NSE announcements and watchlist news, sentiment-scored
08:00	**Analyse**	Previous day’s FII/DII flows, OI build-up, India VIX and today’s regime
08:15	**Draft**	Ranked events, key levels, risks and suggested strategy switches
08:45	**Human review**	Desk head edits and approves the brief
09:00	**Pre-open**	Team trades the plan; the agent has logged every source it used

The agent prepares; humans decide. Read-only tools — it never places orders.

GOVERNANCE

Guardrails for AI on the trading desk

The rules, as of October 2026

- Retail algo framework fully applicable to all brokers since 1 April 2026
- Every algo order carries an exchange-issued Algo ID
- Above 10 orders per second per exchange, the strategy must be registered
- API access only through broker-supervised, static-IP-whitelisted set-ups
- SEBI's AI/ML principles: governance, disclosure, testing, fairness, data privacy

What good desks add

- ✓ Kill switch, tested regularly
- ✓ Hard limits on position, loss and order rate
- ✓ Human approval for any AI-proposed order
- ✓ Model register: owner, version, data, last test
- ✓ Audit log of prompts, tool calls and outputs
- ✓ Drift monitoring with automatic de-risking

Regulations change — check the latest SEBI and NSE circulars before any deployment.

THE TOOLKIT

What participants will build with

Data
Python, pandas, Polars, DuckDB, Parquet, Airflow

Machine learning
scikit-learn, LightGBM, XGBoost, statsmodels

Deep learning
PyTorch, PyTorch Forecasting

Backtesting
vectorbt, Backtrader, your own event engine

GenAI
LLM APIs, prompt templates, JSON schemas, eval sets

RAG and vectors
Embedding models, FAISS, pgvector, Qdrant

Agents
MCP servers, agent frameworks, tool permissions

Production
Docker, MLflow, monitoring, approved broker APIs

Every lab runs in Python on Indian market data — no AI background needed.

OUTCOMES

What participants will be able to do

- Build clean, point-in-time NSE datasets and leak-free features
- Train, walk-forward test and cost-adjust ML classifiers and regime models
- Forecast volatility with LSTMs — and know when simpler models win
- Turn announcements and concalls into tested LLM sentiment features
- Build a cited RAG assistant over SEBI circulars and company filings
- Design a governed multi-agent research desk connected through MCP

CAPSTONE Build, backtest and paper-trade one ML + LLM signal on NSE data — then defend it before a mock risk committee.

Not a black box that prints money.

A disciplined, explainable edge — built by your own team.

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For education only; not investment advice. Trading in securities and derivatives carries risk of loss. Backtested results do not guarantee future performance.